QA MATURITY ASSESSMENT · TMMI · STARTUPS

QA maturity assessment of 10 tech startups, based on TMMi

I designed a self-assessment survey based on TMMi, ran it with ten Chilean tech startups from pre-seed to IPO, and turned the results into stage-specific recommendations and a lean quality framework.

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Master's research, PUC Chile, 2025

> Part of my graduation project for the Master in Information Technology and Management (PUC Chile): "Strengthening software quality practices in Chilean tech startups through the design of a service model". Results are aggregated; no company is identified.

Sample

10
startups assessed

70%
respondents in management or C-level roles

70%
teams of five developers or fewer

50%
startups in growth stage

Main findings

- **Overall maturity between TMMi levels 2 and 3.** 70% of startups sit at level 2 (Managed) or 3 (Defined): processes exist but are not standardised.
- **Measurement and improvement are the weakest areas.** Levels 4 and 5 average 2.8; decisions rely on intuition more than data.
- **Defect management is the strongest area (4.1)**, followed by test planning. Roles and responsibilities, metrics and continuous improvement are the gaps.
- **Quality is mostly reactive.** 50% focus on catching errors or "keeping things from blowing up"; 30% work on systematic prevention.

Maturity by startup stage (average score, 1 to 5)

Stage	Level 2 · Managed	Level 3 · Defined	Level 4 · Measured	Level 5 · Optimised
Pre-seed	4.00	3.00	2.50	1.50
Seed	3.50	3.12	2.00	4.00
Growth	3.75	3.50	3.00	2.40
IPO / consolidation	4.75	5.00	4.00	5.00

Recommendations

Roles and responsibilities

Name a QA owner per squad and formalise the Definition of Ready and Definition of Done.

Metrics (GQM)

A basic dashboard: defects by severity, lead and flow time, minimum coverage and rework percentage.

Infrastructure

Isolated staging environments and basic quality gates in CI/CD.

Quick wins in 4 to 8 weeks

- Acceptance criteria on every story.
- A smoke and regression set for each release.
- A standard bug report template.
- A TMMi baseline to measure progress.

From findings to services: Lean Quality Framework for Startups (LQF-S)

I mapped each stage's main gap to a component of a lean quality framework designed for startups: business-driven quality goals (GQM), a risk-first prioritisation matrix, an agile quality loop inside the sprint, and a maturity roadmap that grows with the company.

Stage	Gap found (data)	LQF-S component	Type of service
Pre-seed	No continuous improvement (level 5 = 1.5)	Lean quality foundations	Diagnosis and quick wins
Pre-seed	Weak test organisation (level 3 = 2.0)	Minimum QA organisation	Lightweight process design
Seed	No quality metrics (level 4 = 2.0)	Essential risk-based metrics	Focused implementation
Seed	Unclear QA roles (level 3 = 2.5)	Roles and ownership	Organisational design
Growth	Reactive improvement (level 5 = 2.4)	Improvement and feedback loops	Optimisation project
Growth	QA integrated but not systematic (level 3 ≈ 3.0)	QA–Dev–Business integration	Incremental implementation
IPO	Need for standards and benchmarks (levels 4–5 high)	Scaling and quality governance	Advanced consulting